

FEATURE IMPORTANCE ANALYSIS

I used the Select Attributes tab in WEKA to rank all 20 attributes by their information gain value. Information gain is used to determine the usefulness of each attribute to predict the target class, which is the student performance group. The results were very clear:

Attendance was the top attribute with an information gain of 0.3557, more than double that of the second ranked attribute Hours_Studied at 0.1514. Physical_Activity and Sleep_Hours both had a score of 0, which indicates that they are not useful for this classification problem at all.

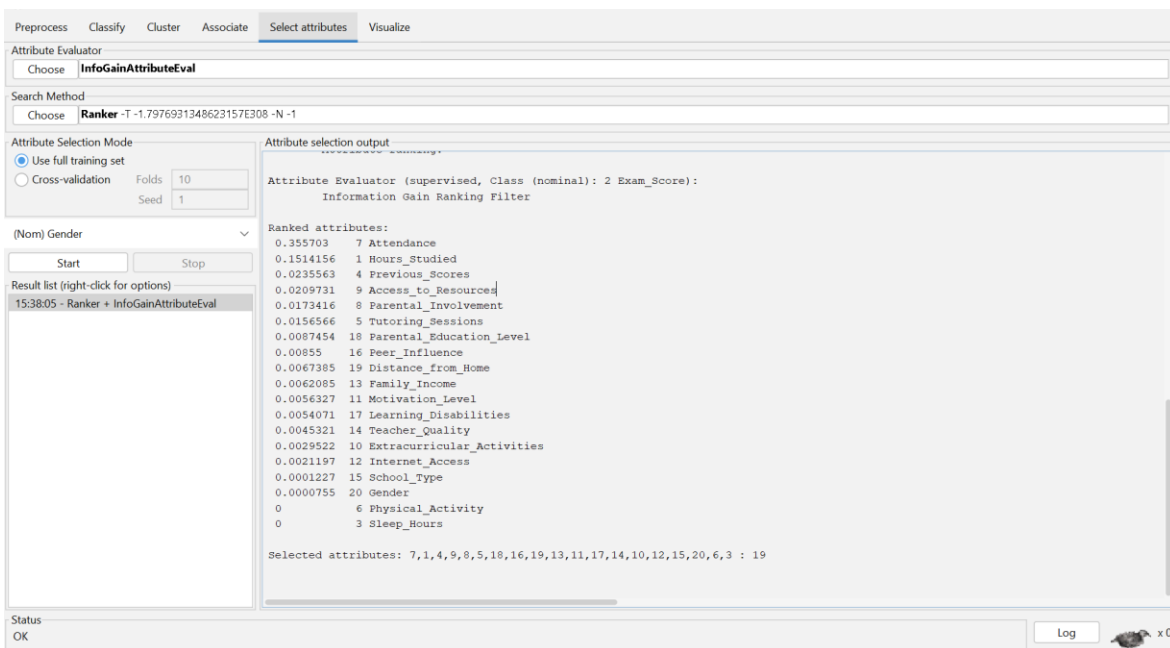


Figure 2 : WEKA Select Attributes Tab InfoGain Ranking using InfoGainAttributeEval and Ranker. The information gain of Attendance is 0.3557, which is the most important predictor. Hours_Studied was ranked second and Previous_Scores was ranked third. If the info